DEEP LEARNING · FINAL PROJECT · 2026

# Reading the Khmer province on Cambodian licence plates

Five deep-learning approaches compared

26 CLASSES | 5 APPROACHES | 567 TEST CROPS




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# The real-world problem: why province matters

01 · SMART ACCESS

## Automated Entry

Boreys, parking & highway tolls in Cambodia need fast, hands-free vehicle verification.

02 · FATAL FLAW

## Duplicate Numbers

`2A-0243` repeats across 26 provinces. Number alone causes false gate access!

03 · OCR LIMITATION

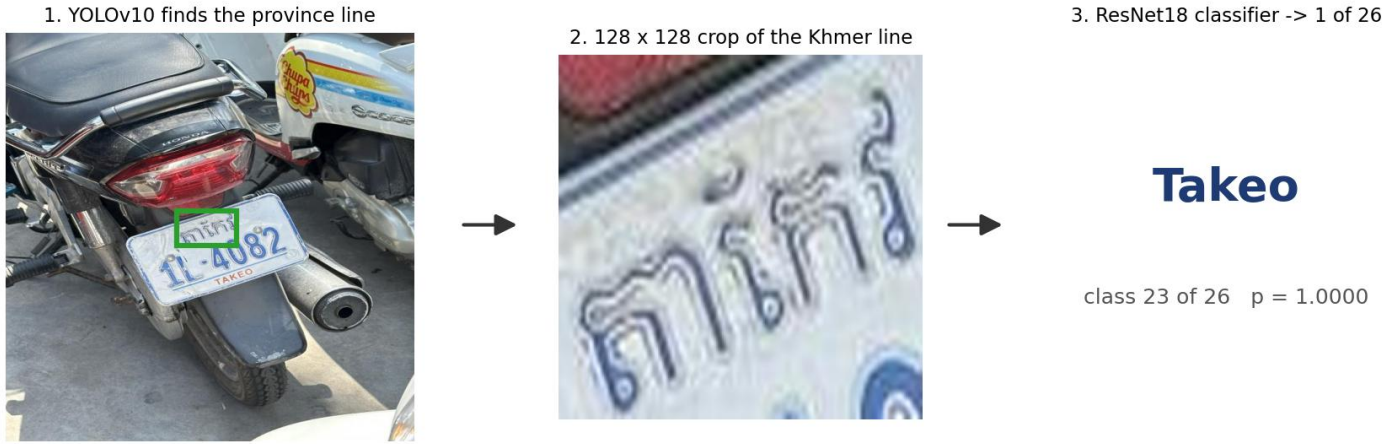
## Commercial OCR Fails

Standard engines fail on Khmer script: complex stacked ligatures break character OCR.

04 · DEEP LEARNING

## Whole-Line Classifier

Treat the top line as an image: **128×128 crop → 1 of 26 classes** with deep networks.



# End-to-end ALPR system design

**01 · INPUT**

## Camera Stream

1080p Video Stream

**LIVE FEED**
30 FPS Video
Road & Barrier
Gate Monitoring

Raw Vehicle Frame

**02 · DETECTION**

## YOLOv10 Detector

Anchor-free Plate BBox

**LOCALIZATION**
Latency: ~15 ms
143/149 Recall
Tight Crop Box

Cropped Plate BBox

**THIS STUDY · 26 CLASSES**

## Khmer Province Classifier

ViT-B/16 & ResNet18 (128x128)

Province: "Takeo" (96.1 %)

**03B · LATIN & DIGITS**

## CRNN + CTC Number Reader

CNN + BiLSTM + CTC Decoder

Number: "1L-4082" (83.9 %)

**04 · FUSION**

## Plate Fusion

Canonical Formatter

**STRING FUSION**
Combine Lines
Syntax Check
Latency: < 1 ms

Takeo 1L-4082

**05 · ACCESS**

## Security Gate

SQLite Whitelist

✓ ALLOW
Gate Opens

✕ DENY
Remain Closed

0 False Accepts

# Plate_v4 — Roboflow, CC BY 4.0

| 2 515 | 529 | 567 | 26 | 13.4 % |
|---|---|---|---|---|
| train crops | validation | test | classes | test near-duplicates |

1 **Public benchmark dataset**
Roboflow Plate_v4 (CC BY 4.0): 3 611 vehicle crops across real Cambodian road conditions.

2 **13.4 % near-duplicate leakage**
76 of 567 test crops share source photos with training. All models also evaluated on clean n=491.

3 **Standardized crop inputs**
128×128 (CNN) / 224×224 (ViT) patches focusing exclusively on the top Khmer province line.

4 **Severe class imbalance (25:1)**
*Other* (453) vs. *Mondul Kiri* (18). 4 rare provinces have ≤ 50 crops, requiring **macro-F1**.

# Class distribution — 26 provinces

| PROVINCE | TRAIN | TEST | PROVINCE | TRAIN | TEST |
|---|---|---|---|---|---|
| Other | 453 | 98 | Kampong Chhnang | 70 | 16 |
| Phnom Penh | 254 | 55 | Kampot | 70 | 15 |
| Preah Sihanouk | 148 | 33 | Pursat | 67 | 16 |
| Siem Reap | 147 | 32 | Preah Vihear | 65 | 15 |
| Kampong Cham | 110 | 25 | Pailin | 64 | 15 |
| Prey Veng | 101 | 23 | Kratie | 63 | 15 |
| Takeo | 100 | 23 | Kampong Thom | 62 | 14 |
| Kandal | 97 | 22 | Tboung Khmum | 61 | 14 |
| Battambang | 87 | 20 | Stung Treng | 52 | 12 |
| Svay Rieng | 86 | 20 | Koh Kong | 47 | 11 |
| Banteay Meanchey | 82 | 19 | Kep | 42 | 9 |
| Kampong Speu | 77 | 18 | Ratanakiri | 20 | 5 |
| Oudor Meanchey | 72 | 17 | Mondul Kiri | 18 | 5 |

# Identical setup for every approach

## TRAINING

1. 128×128 (CNN) / 224×224 (ViT)
2. Crop / affine / colour-jitter augmentation
3. 40 epochs · batch 32 · Adam · StepLR

**Seed 42**
Python · NumPy · PyTorch · cuDNN · DataLoader

## EVALUATION

1. Model selected on **validation** only
2. Test set touched **once** per run
3. Accuracy + **macro-F1** (imbalance)

**Colab T4**
≈ 6–8 minutes per run

# The five approaches

| | ARCHITECTURE | INITIAL WEIGHTS | TRAINING STRATEGY | PARAMETERS |
|---|---|---|---|---|
| A | ResNet18 | Random (scratch) | All layers trainable | 11.2 M |
| B | ResNet18 | ImageNet pre-trained | Final FC only (backbone frozen) | 13 k (0.12 %) |
| C | ResNet18 | ImageNet pre-trained | Full fine-tuning (low learning rate) | 11.2 M |
| D | Small 4-block CNN | Random (scratch) | All layers trainable | 0.4 M |
| E | ViT-B/16 (Transformer) | ImageNet pre-trained | Full fine-tuning (196 patches) | 85.8 M |

01 · PRE-TRAINING

**A** (scratch) vs **C** (fine-tune)

02 · FEATURE TRANSFER

**B** (frozen) vs **C** (fine-tune)

03 · MODEL CAPACITY

**D** (0.4M CNN) vs **A** (11.2M)

04 · ATTENTION VS CONV

**E** (ViT-B/16) vs **C** (ResNet18)

# Scored on the same 567 test crops

| | APPROACH | VAL ACC | TEST ACC | CLEAN (N=491) | MACRO-F1 | KEY OUTCOME |
|---|---|---|---|---|---|---|
| A | ResNet18 scratch | 93.8 % | 93.3 % | 94.7 % | 92.9 % | Strong baseline (2 515 imgs) |
| B | ResNet18 frozen | 40.1 % | 42.0 % | 43.2 % | 37.9 % | Domain shift collapse |
| C | ResNet18 fine-tune | 92.6 % | 94.0 % | 95.3 % | 92.4 % | Fast 5-epoch convergence |
| D | Small 4-block CNN | 48.4 % | 46.6 % | 47.7 % | 40.7 % | Under-capacity (0.4M) |
| E | ViT-B/16 (Transformer) | 96.2 % | 96.1 % | 96.5 % | 95.8 % | Top accuracy + macro-F1 |

FINDING 1 · PRE-TRAINING

**A ≈ C · p = 0.63**

Pre-trained weights didn't make the model smarter they just made it learn faster.

FINDING 2 · FEATURE TRANSFER

**B Collapses to 42.0 %**

ImageNet knows dogs and cars, not Khmer letters. We must retrain the whole network so it can learn Khmer strokes.

FINDING 3 · MODEL CAPACITY

**D Fails at 46.6 %**

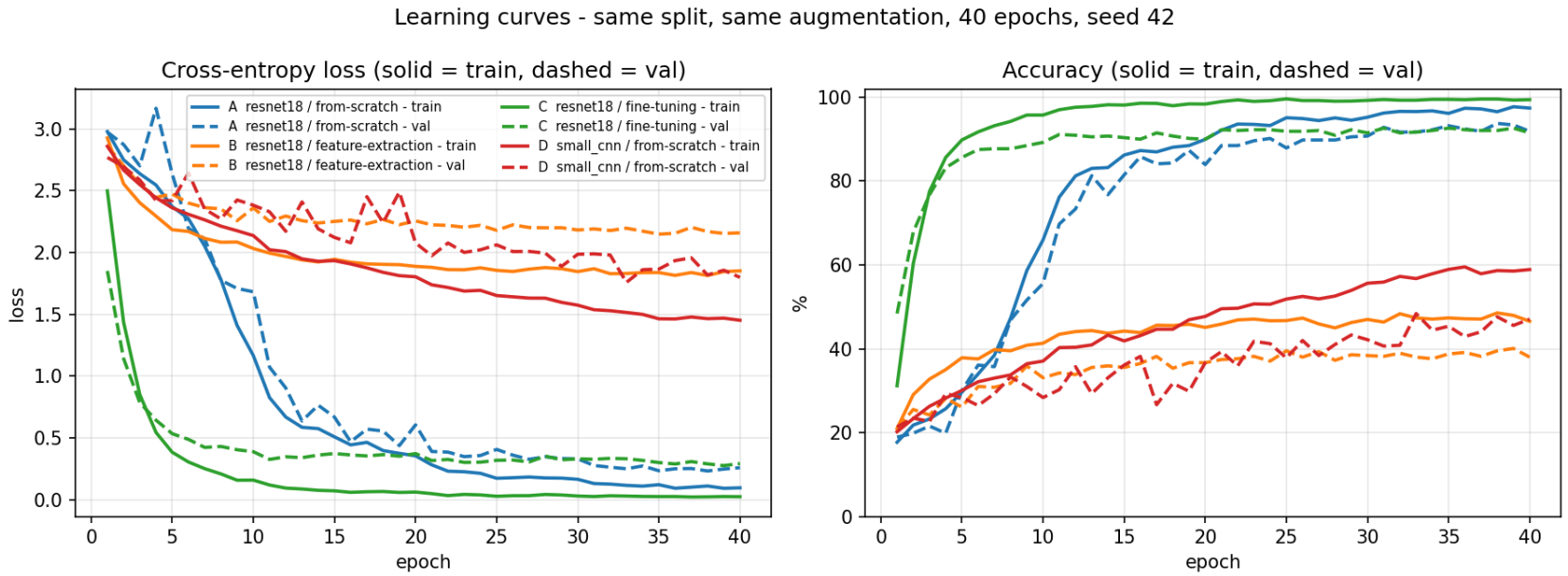
A small 400k model is too simple. Khmer words are complex, so we need a deeper model with more capacity.

FINDING 4 · TRANSFORMERS

**ViT Wins: 96.1 % / 95.8 %**

The Vision Transformer is our best model because its attention mechanism looks at the entire Khmer word at once.

# Learning curves




**5 vs 20**
epochs to 85 % — C vs A

**97–99 %**
train acc A, C — mild over-fit

**< 60 %**
train acc B, D — under-fit

# Tuning C — selected on validation

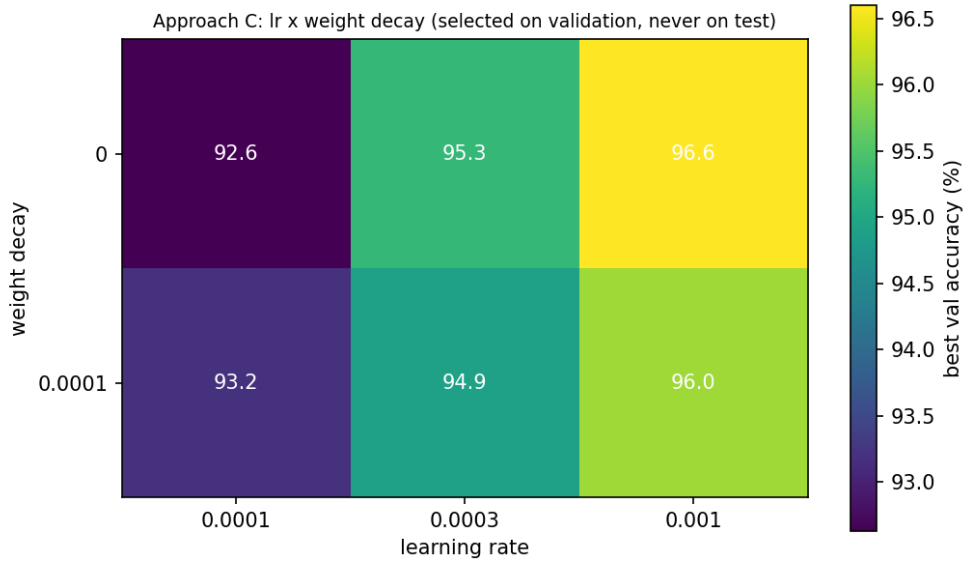
| LEARNING RATE | WD = 0 | WD = 1E-4 |
|---|---|---|
| 1e-4 | 92.6 % | 93.2 % |
| 3e-4 | 95.3 % | 94.9 % |
| **1e-3** | **96.6 % ← selected** | 96.0 % |

1. Textbook "low LR for fine-tuning" was the **worst** cell
2. 10× larger LR → **+4 points** validation accuracy
3. Weight decay: no help — augmentation already regularises

**96.5 %**

final model — test accuracy · lr 1e-3, wd 0

VALIDATION ACCURACY · LR × WEIGHT DECAY



# Where the 34 errors come from

**8 of 34**

one pair — Banteay ↔ Oudor Meanchey

1. បៈមានជ័យ vs ឱៈមានជ័យ — **one glyph** apart
2. Other errors also share a whole word
3. Rest: blur, 20-px text, JPEG artefacts
4. Smallest classes **90 %** vs 94.2 %

MOST CONFIDENT WRONG PREDICTIONS · C



# Why — in course concepts

| | | |
|---|---|---|
| B → 42 % | DOMAIN SHIFT | ImageNet filters are not Khmer strokes |
| A ≈ C | DATA SIZE | 2 515 images suffice from random init |
| D → 47 % | CAPACITY | 0.4 M params cannot separate 26 words |
| E → 96.1 % | ATTENTION | global self-attention captures full glyph ligatures |
| tuning +2.5 | OPTIMISATION | fine-tuning wants a training-size LR |

# What this study cannot claim

1. Public scraped data — not purpose-collected
2. **13.4 %** near-duplicates in the test split
3. 567 crops → **± 2 points**; A vs C not separable
4. One seed per configuration
5. CNNs and Vision Transformers evaluated (2 architecture families)

# Four takeaways

01

**Train the whole network** — frozen features do not transfer (42 %)

02

**Transformers excel on script** — ViT matches 96.1 % & lifts upside-down to 26.8 %

03

**Capacity matters** — 0.4 M parameters is not enough; 11 M to 86 M is

04

For CNNs, biggest gain came from **tuning the learning rate**

**Next:** re-split by source photo · photograph the five smallest provinces · benchmark Swin Transformer & edge quantization

QUESTIONS WELCOME

# Thank you

Every number came from a script in the repository.

**github.com/SoSereysokbotra**



APPENDIX

# Supporting material




full results · statistics · tuning grid · the gate system · confusion · per-class · measurement · code

# Full results — every column

| | APPROACH | VAL | TEST | CLEAN N=491 | MACRO-F1 | TRAINABLE | TIME |
|---|---|---|---|---|---|---|---|
| A | ResNet18 scratch | 93.8 % | **93.3 %** | 94.7 % | 92.9 % | 11 189 850 | 6.1 min |
| B | ResNet18 frozen | 40.1 % | **42.0 %** | 43.2 % | 37.9 % | 13 338 | 5.3 min |
| C | ResNet18 fine-tune | 92.6 % | **94.0 %** | 95.3 % | 92.4 % | 11 189 850 | 5.9 min |
| D | small CNN | 48.4 % | **46.6 %** | 47.7 % | 40.7 % | 396 058 | 5.3 min |
| E | ViT-B/16 fine-tune | 96.2 % | **96.1 %** | 96.5 % | 95.8 % | 85 818 650 | 8.0 min |

1. Model chosen on **val**; test read **once** per run
2. Clean subset = the 491 crops with no shared source photo — accuracy is *higher*, so the leak did not inflate anything
3. All runs: 40 epochs, batch 32, seed 42, Colab T4

# Is the A–C difference real?

**McNemar exact test**, paired on the same 567 crops

| | C RIGHT | C WRONG |
|---|---|---|
| A right | 512 | 17 |
| A wrong | 21 | 17 |

1. 38 discordant pairs → **p = 0.627**
2. **Not significant** — A and C are a tie
3. I do not claim a winner between them

**95 % Wilson confidence intervals** (n = 567)

| APPROACH | TEST | 95 % CI | WIDTH |
|---|---|---|---|
| A | 93.3 % | 90.9 – 95.1 | 4.1 pts |
| B | 42.0 % | 38.0 – 46.1 | 8.1 pts |
| C | 94.0 % | 91.7 – 95.7 | 3.9 pts |
| D | 46.6 % | 42.5 – 50.7 | 8.2 pts |
| E | 96.1 % | 94.2 – 97.4 | 3.2 pts |

A, C and E overlap in high confidence — which is why **567 images cannot separate top models.**

# Full tuning grid — all six runs

| LR | WEIGHT DECAY | BEST EPOCH | VALIDATION | TEST | MACRO-F1 |
|---|---|---|---|---|---|
| **1e-3** | **0** | 33 | **96.60 % ← selected** | 96.47 % | 96.04 % |
| 1e-3 | 1e-4 | 35 | 96.03 % | 95.41 % | 95.06 % |
| 3e-4 | 0 | 15 | 95.27 % | **97.00 %** | 96.55 % |
| 3e-4 | 1e-4 | 27 | 94.90 % | 95.94 % | 95.35 % |
| 1e-4 | 1e-4 | 36 | 93.19 % | 94.53 % | 93.02 % |
| 1e-4 | 0 | 31 | 92.63 % | 94.00 % | 92.44 % |

! **Yes — 3e-4 scores higher on test (97.0) than the model I selected.** Selecting it *because* it won on test would be selecting on the test set — the exact mistake this project is about avoiding. Validation chooses; test is reported once.

# The ALPR system this model serves

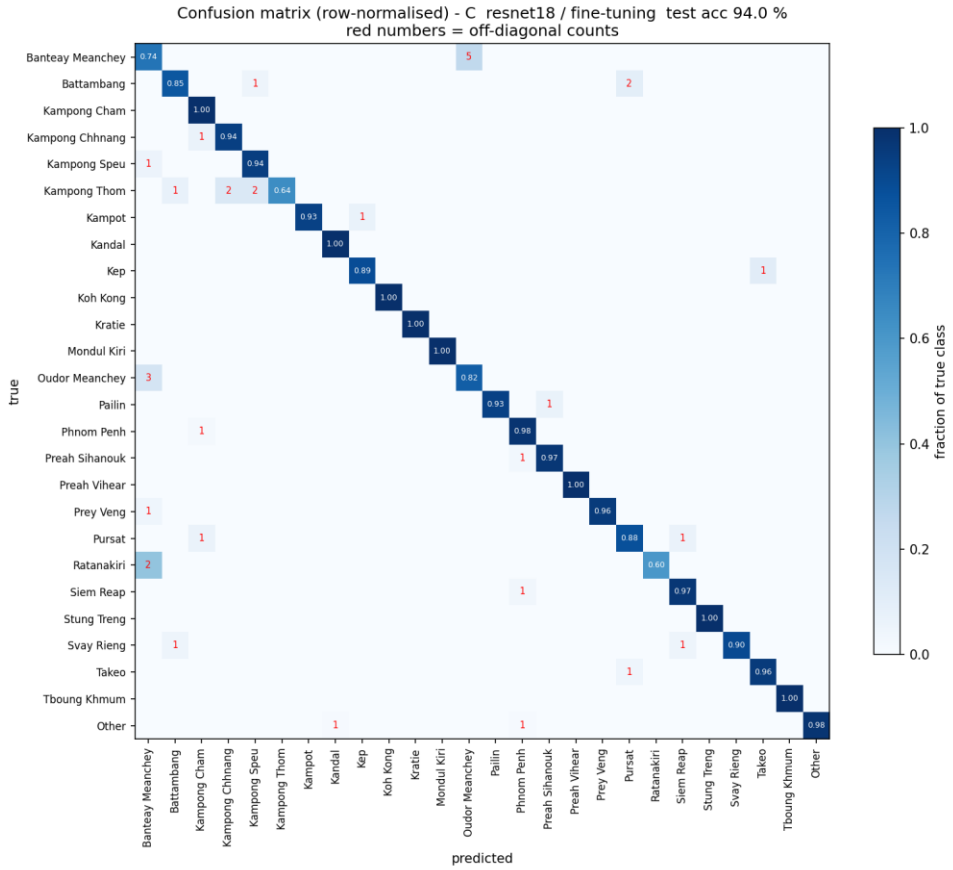



## END-TO-END ON REAL SCENES

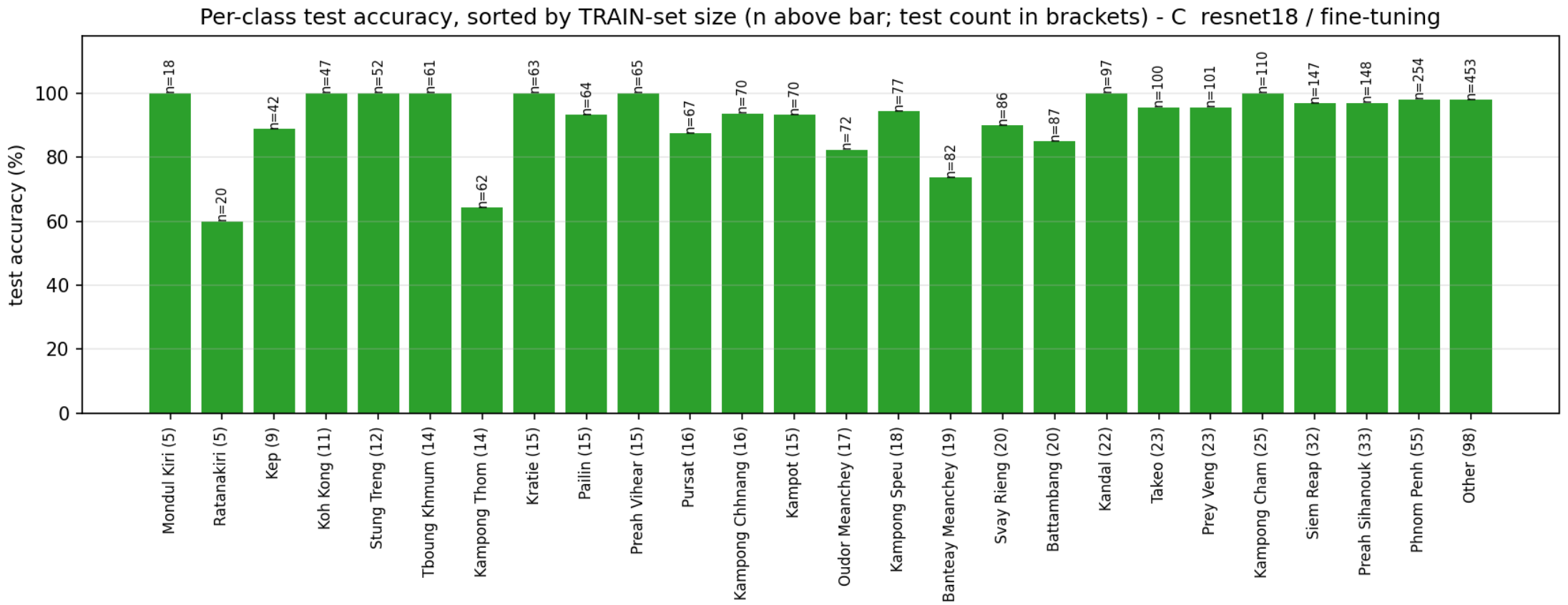
| 143/149 | 83.9 % | 94.0 % | 0 | ~50 ms |
|---|---|---|---|---|
| plate detected | number correct | province correct | false accepts | per frame |

149 labelled real scenes · RTX 3050 · whitelist of 137 near-miss plates

# Confusion matrix — approach C



# Per-class accuracy vs training-set size



# Measurement discipline

01

CRNN test set: **34.9 %** plate-level leakage found

02

Headline fell 69.1 → **51.7 %** — the lower number is reported

03

Spatial Transformer proved to do **nothing** — negative result

04

`experiment_log.csv` — every measurement tagged with its commit

Laroca et al. (IJCNN 2023) report 19–68 % duplicates in standard LPR benchmarks.

# The code behind B and D

**Feature extraction (B)** — only the head trains

```
for name, prm in model.named_parameters():
    if not name.startswith('fc.'):
        prm.requires_grad = False      # 11 176 512 frozen
    else:
        trainable += prm.numel()       # 13 338 trainable

optimizer = Adam(
    [p for p in model.parameters() if p.requires_grad],
    lr=args.lr, weight_decay=args.weight_decay)
```

**SmallCNN (D)** — 4 blocks, 128→64→32→16→8, then GAP

```
def block(cin, cout):
    return nn.Sequential(
        nn.Conv2d(cin, cout, 3, padding=1),
        nn.BatchNorm2d(cout), nn.ReLU(True),
        nn.MaxPool2d(2))

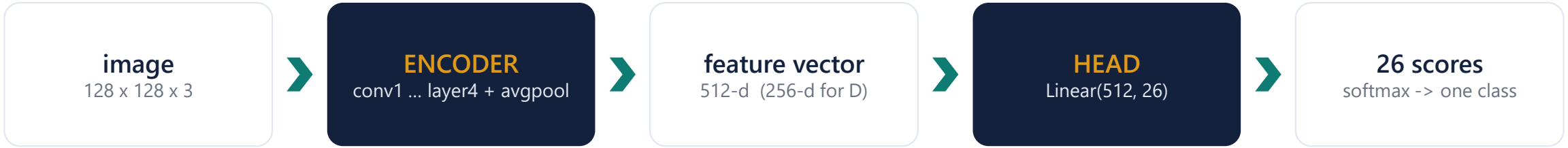
self.features = nn.Sequential(
    block(3,32), block(32,64), block(64,128), block(128,256))
self.pool = nn.AdaptiveAvgPool2d(1)   # 396 058 params total
```

**Seed 42** in Python, NumPy, PyTorch, cuDNN and the DataLoader

`last.pth` saved every epoch + `--resume` for Colab disconnects

The lr 1e-4 grid cell reproduced run C **to the digit**

# Encoder, head — and why there is no decoder




| APPROACH | ENCODER | FEATURE | HEAD | DECODER |
|---|---|---|---|---|
| A / B / C | ResNet18 conv stack | 512-d | Linear(512, 26) | none |
| D | 4 conv blocks + GAP | 256-d | Linear(256, 26) | none |
| E | PatchEmbed + 12 Transformer blocks | 768-d (CLS) | Linear(768, 26) | none |

1 A **decoder generates a sequence**, one step at a time, each step conditioned on the previous one

2 My output is **one label out of 26** — there is nothing to generate, so a decoder would be architecturally meaningless

! `CTCDecoder` in the number reader is **not a neural decoder** — it is a collapse rule: argmax per timestep, merge repeats, drop blanks. **No weights, never trained.**